

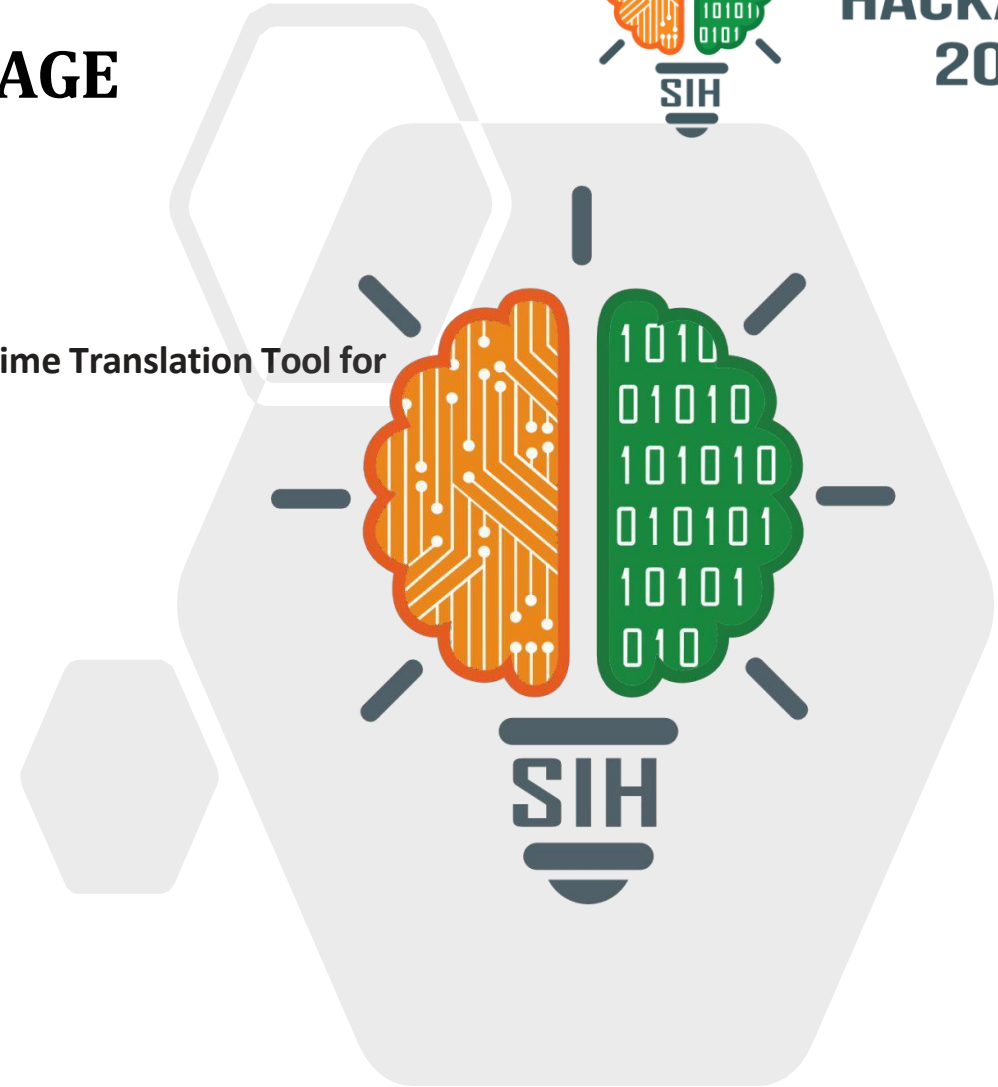
SMART INDIA HACKATHON 2026

TITLE PAGE

- Problem Statement ID – SIH26042
- Problem Statement Title – AI-Powered Vernacular Pedagogy and Real-Time Translation Tool for Mother Tongue-Based Primary Education
- Theme – Smart Education
- PS Category – Software
- Team ID –
- Team Name (Registered on portal) – Loading...



**SMART INDIA
HACKATHON
2026**



Curriculum-Synced Classroom Translation Companion for Ho, Mundari & Santhali Linguistic Tribes

THE PROBLEM

- Only 23% of Grade-1 children in Jharkhand's tribal-area schools understand Hindi (PALASH language mapping survey, 2024) — most speak Ho, Mundari or Santhali at home
- 5,000+ tribal-area schools need MTB-MLE, but Hindi-medium-trained teachers lack tribal-language proficiency — the core bottleneck blocking PALASH from scaling
- General-purpose tools (e.g. Adi Vaani) solve one-off phrase translation — not real-time, curriculum-synced classroom teaching dialogue

OUR SOLUTION

- AI-assisted translation & curriculum companion built for the PALASH FLN classroom — not a general translator
- Curriculum-bounded Hindi $\leftrightarrow$ Ho/Mundari/Santhali phrase-matching engine mapped directly to NIPUN Bharat FLN content
- Real-time two-way voice dialogue between teacher and student, sub-3-second latency, fully offline on ≤ 2 GB RAM tablets
- Auto-generated bilingual worksheets & visual flashcards aligned to FLN learning outcomes

INNOVATION & UNIQUENESS

- Adi Vaani solves general-purpose translation; we solve classroom-embedded, curriculum-synced teaching dialogue — a narrower, deployable-first problem
- Honest, curriculum-bounded scope (~2,000–3,000 FLN words) makes accurate low-resource translation realistic today, instead of overpromising open-domain MT

TECHNOLOGY STACK

FRONTEND — Android (Kotlin, Offline-First)

Teacher Dashboard · Live Voice Companion · Worksheet/Flashcard Generator

ON-DEVICE AI ENGINE — TF Lite / ONNX (Quantized)

ASR: IndicWav2Vec/Vosk · Curriculum-Bounded Match & Retrieval Engine · TTS: Vakyansh/VITS

LOCAL STORAGE

SQLite — Curriculum Cache, FLN Phrase Bank & Translation Memory

BACKEND — Cloud (periodic sync only)

Node.js/FastAPI API Gateway · PostgreSQL Curriculum & Phrase-Bank DB · Update Server

DATA SOURCES & SECURITY

PALASH FLN Curriculum + AI4Bharat/Bhashini (bootstrap & validation) · Teacher feedback loop · JWT + AES-256

CLASSROOM INTERACTION FLOW (TWO-WAY, WITH FALLBACK)

Teacher speaks Hindi



ASR captures & normalizes speech



Match against FLN phrase bank



Match found → play tribal-language audio to student



Student responds → reverse match → Hindi output to teacher

No match → flagged & logged → added to phrase-bank expansion queue for teacher review

FEASIBILITY

- ✓ PALASH already has curated Hindi FLN curriculum content ready to seed the phrase bank
- ✓ FLN vocabulary is domain-bounded (~2,000–3,000 words) — a realistic scope for accurate curation, unlike open-domain MT
- ✓ AI4Bharat / Bhashini provide reference models & tooling to bootstrap and validate the pipeline, not a dependency for full fine-tuning
- ✓ Retrieval-based matching (not generative inference) keeps latency low and works fully offline on ≤ 2 GB RAM devices

CHALLENGES & RISKS

- ❓ Very limited existing digital parallel corpora for Ho and Mundari specifically
- ❓ Dialect variation within a single tribal language
- ❓ Teacher trust/adoption of AI-suggested translations
- ❓ Intermittent connectivity for the initial content sync in remote schools

MITIGATION STRATEGIES

- Start with Santhali (Ol Chiki script, strongest existing resource base), then expand to Ho & Mundari
- Partner with PALASH linguists & native speakers for structured validation sprints to build the seed phrase bank
- Human-in-the-loop design — teachers approve/correct translations before use, feeding continuous improvement
- Offline-first architecture with Wi-Fi hotspot / SD-card sync fallback for zero-connectivity schools

IMPACT AND BENEFITS

Target Audience: Teachers & students across 5,000+ tribal-area primary schools in Jharkhand under the PALASH MTB-MLE programme

5,000+

tribal-area primary
schools reachable

3

languages addressed
(Ho, Mundari, Santhali)

<3 sec

voice translation
latency, fully offline

Educational

Enables genuine mother-tongue instruction; improves foundational literacy & numeracy outcomes aligned to NIPUN Bharat targets.

Social

Removes the teacher-language bottleneck blocking MTB-MLE at scale; children learn in the language they understand at home.

Economic

Cuts the cost and time of recruiting/training native tribal-language teachers — technology bridges the gap.

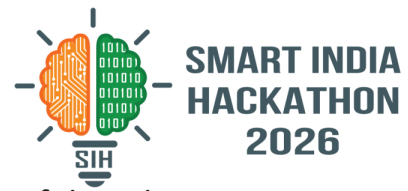
Scalability

Retrieval-based framework is language-agnostic in design — extensible to other underserved mother tongues beyond Jharkhand.

POLICY ALIGNMENT

Directly supports the NIPUN Bharat Mission and NEP 2020 mandate for mother-tongue / home-language-based foundational education, and strengthens the state's MTB-MLE (PALASH) rollout at scale without requiring large-scale hiring of tribal-language teachers.

RESEARCH AND REFERENCES



- **PALASH Programme** – Dept. of School Education & Literacy, Govt. of Jharkhand — MTB-MLE case study featured in UNESCO's State of the Education Report for India 2025
- **NIPUN Bharat Mission** – Ministry of Education, Govt. of India — Foundational Literacy & Numeracy (FLN) guidelines
- **AI4Bharat** – IndicTrans2, IndicWav2Vec, Vakyansh TTS — open Indic-language NLP models — ai4bharat.iitm.ac.in
- **Digital India Bhashini Mission** – National multilingual AI translation platform — bhashini.gov.in
- **Adi Vaani** – Ministry of Tribal Affairs' AI tribal-language translator (IIT Delhi-led consortium) — reference point for differentiation: general-purpose translation vs. our classroom-embedded, curriculum-synced approach
- **NCERT** – Foundational Literacy and Numeracy (FLN) curriculum framework